

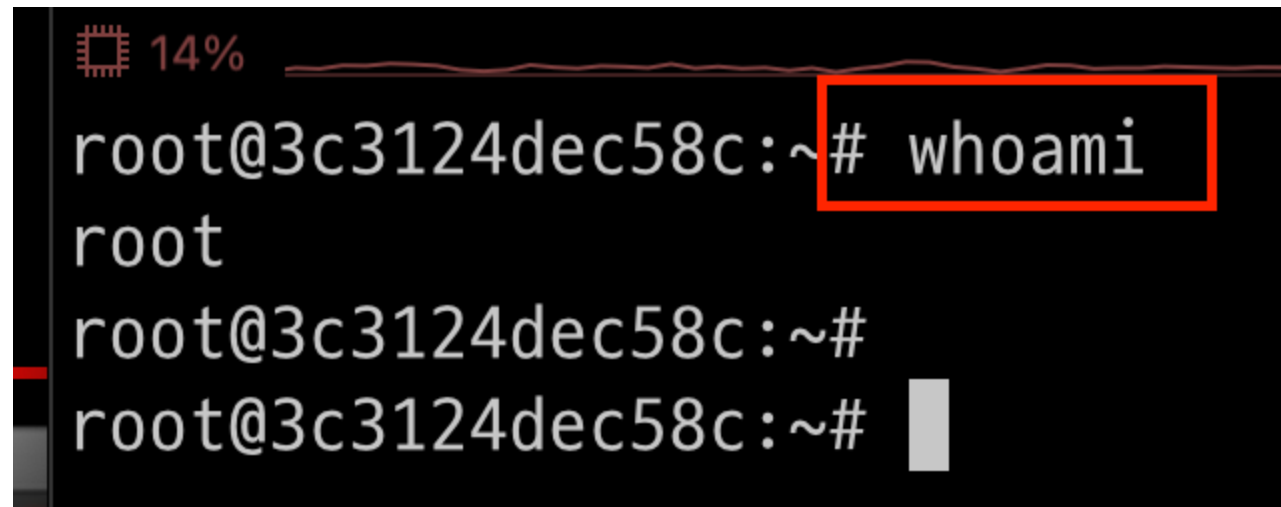
Linux Account

- LINUX는 사용자 중심의 시스템으로, 사용자 계정이 있어야 접속할 수 있다.
- root
 - 시스템을 관리할 수 있는 관리자 권한의 계정이자 슈퍼 유저
 - 리눅스 파일 체제의 최상위 디렉토리(/)로도 표현한다.
- 일반 사용자 계정
 - root가 useradd 명령어를 통해 생성한 모든 계정
 - root와 달리 권한이 있는 파일과 디렉토리에 대해서만 읽고 쓸 수 있다.

Linux Account 명령어

계정확인

whoami



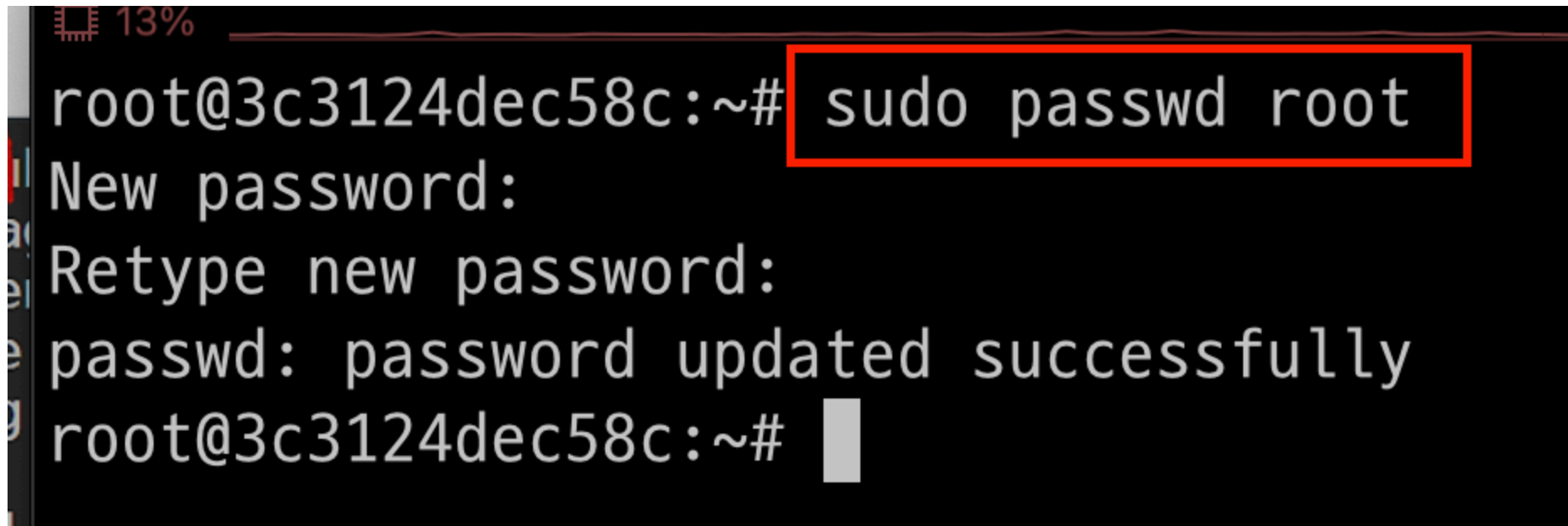
```
❏ 14%  
root@3c3124dec58c:~# whoami  
root  
root@3c3124dec58c:~#  
root@3c3124dec58c:~#
```

A terminal window with a black background. At the top left, there is a red icon of a microchip followed by the text '14%'. The terminal shows a prompt 'root@3c3124dec58c:~#' followed by the command 'whoami' which is enclosed in a red rectangular box. The output of the command is 'root'. Below this, the prompt appears again twice, indicating the command was executed and the prompt returned.

root > 비번 생성(변경)

- `sudo` : 사용자 계정에 root권한을 일시적으로 부여하는 방법
- sudo 오류 발생시 다음 페이지 참고

```
sudo passwd root
```

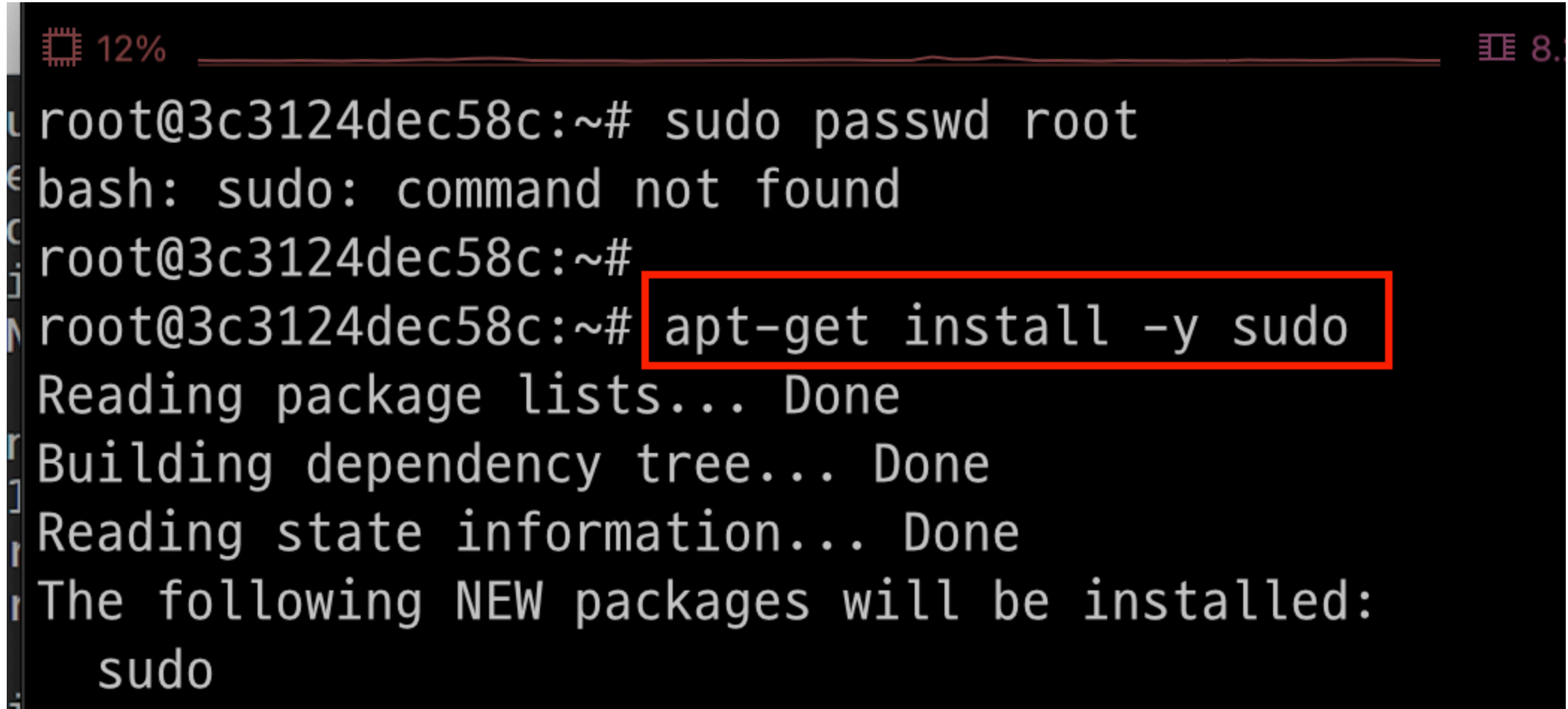


A terminal window screenshot with a black background and white text. The prompt is 'root@3c3124dec58c:~#'. The command 'sudo passwd root' is entered and highlighted with a red rectangular box. The output shows the password update process: 'New password:', 'Retype new password:', and 'passwd: password updated successfully'. The prompt returns to 'root@3c3124dec58c:~#' with a cursor. In the top left corner, there is a battery icon and '13%'.

```
root@3c3124dec58c:~# sudo passwd root
New password:
Retype new password:
passwd: password updated successfully
root@3c3124dec58c:~#
```

- sudo 오류 발생 > 처리 후 다시 passwd 설정

```
apt-get install -y sudo
```

A terminal window with a dark background and light-colored text. The terminal shows a sequence of commands and their outputs. The first command is 'sudo passwd root', which results in 'bash: sudo: command not found'. The second command is 'apt-get install -y sudo', which is highlighted with a red rectangular box. The output of this command shows the progress of installing the 'sudo' package, including reading package lists, building a dependency tree, and reading state information. The final output indicates that the 'sudo' package will be installed.

```
12% 8.
root@3c3124dec58c:~# sudo passwd root
bash: sudo: command not found
root@3c3124dec58c:~#
root@3c3124dec58c:~# apt-get install -y sudo
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
sudo
```

계정 생성

- 명령어: adduser <생성될 계정아이디>

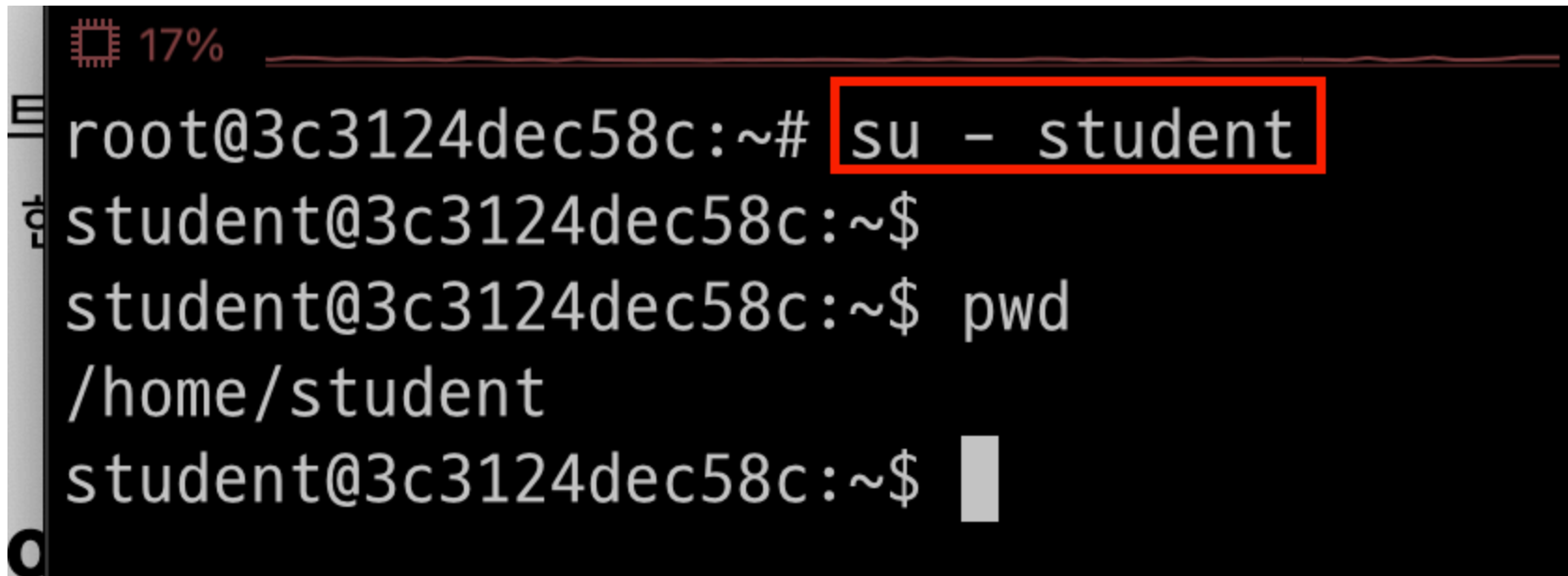
```
adduser student
```

```
root@3c3124dec58c:~# adduser student
Adding user `student' ...
Adding new group `student' (1000) ...
Adding new user `student' (1000) with group `student' ...
Creating home directory `/home/student' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
root@3c3124dec58c:~#
```

계정 변경

- 명령어: su - <계정명>

```
su - student  
pwd # student의 홈 디렉토리임
```



A terminal window with a black background and white text. At the top left, there is a battery icon and '17%'. The prompt is 'root@3c3124dec58c:~#'. The command 'su - student' is entered and highlighted with a red rectangular box. The prompt changes to 'student@3c3124dec58c:~\$'. The command 'pwd' is entered, and the output is '/home/student'. The prompt returns to 'student@3c3124dec58c:~\$' with a cursor at the end.

```
root@3c3124dec58c:~# su - student  
student@3c3124dec58c:~$  
student@3c3124dec58c:~$ pwd  
/home/student  
student@3c3124dec58c:~$
```

student > 디렉토리 및 파일 생성

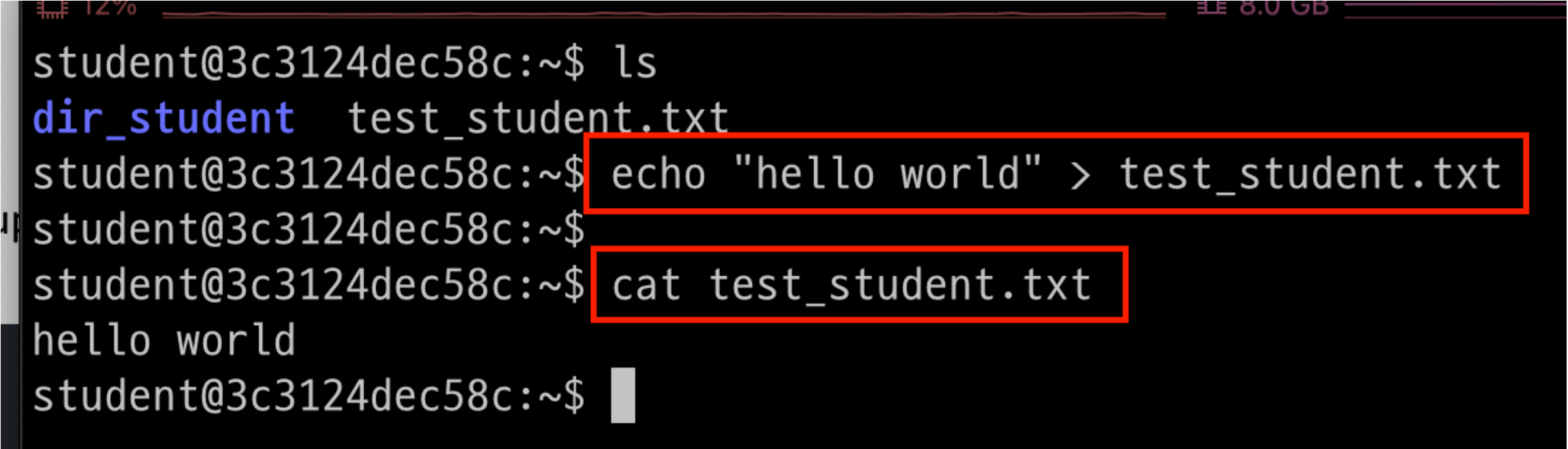
- test_student.txt 은 student 사용자
와 그룹만 수정이 가능함

```
touch test_student.txt
mkdir dir_student
ls -al # 권한 확인
```

```
student@3c3124dec58c:~$ touch test_student.txt
student@3c3124dec58c:~$ mkdir dir_student
student@3c3124dec58c:~$
student@3c3124dec58c:~$ ls -al
total 24
drwxr-x--- 3 student student 4096 Feb  5 05:32 .
drwxr-xr-x 1 root    root    4096 Feb  5 05:21 ..
-rw-r--r-- 1 student student  220 Feb  5 05:21 .bash_logout
-rw-r--r-- 1 student student 3771 Feb  5 05:21 .bashrc
-rw-r--r-- 1 student student  807 Feb  5 05:21 .profile
drwxrwxr-x 2 student student 4096 Feb  5 05:32 dir_student
-rw-rw-r-- 1 student student   0 Feb  5 05:32 test_student.txt
student@3c3124dec58c:~$
```

student > 파일 수정 및 읽기

```
# 파일 수정
echo "hello world" > test_student.txt
# 파일 읽기
cat test_student.txt
```

A terminal window with a dark background and light-colored text. The prompt is 'student@3c3124dec58c:~\$'. The first command is 'ls', which lists 'dir_student' and 'test_student.txt'. The second command, 'echo "hello world" > test_student.txt', is highlighted with a red rectangular box. The third command, 'cat test_student.txt', is also highlighted with a red rectangular box. The output of the cat command is 'hello world'. The prompt is now 'student@3c3124dec58c:~\$' with a cursor. At the top of the terminal, there are status bars showing '12%' and '8.0 GB'.

```
student@3c3124dec58c:~$ ls
dir_student  test_student.txt
student@3c3124dec58c:~$ echo "hello world" > test_student.txt
student@3c3124dec58c:~$
student@3c3124dec58c:~$ cat test_student.txt
hello world
student@3c3124dec58c:~$
```

student > sudo 사용하기

- sudo 권한이 없다는 오류 발생

```
sudo apt-get install python3 python-pip3
```

15%

8.0 GB

```
student@3c3124dec58c:~$ sudo apt-get install python-pip3
```

```
[sudo] password for student:
```

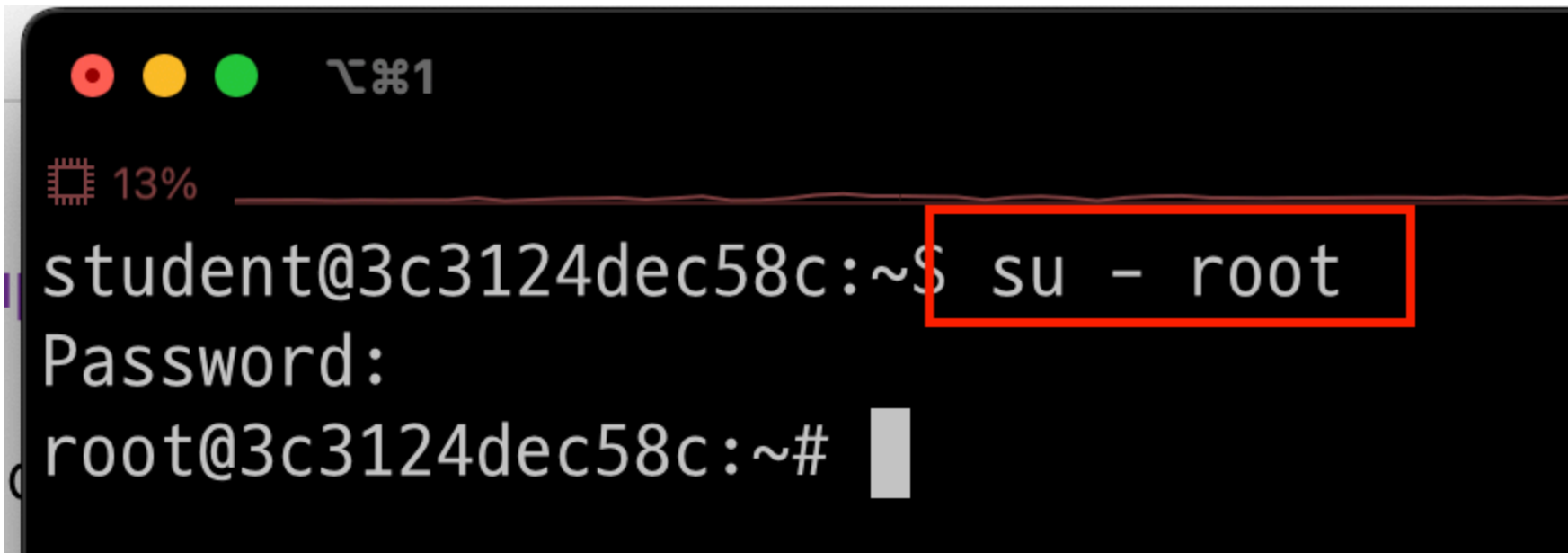
```
student is not in the sudoers file. This incident will be reported.
```

```
student@3c3124dec58c:~$
```

```
student@3c3124dec58c:~$
```

- root 계정으로 변경
- 명령어: su - <계정명>

```
su - root
```



A terminal window with a dark background. At the top, there are three colored window control buttons (red, yellow, green) and a keyboard shortcut icon. Below them is a battery status icon and '13%'. The prompt is 'student@3c3124dec58c:~\$'. The command 'su - root' is entered and highlighted with a red rectangular box. Below the command, the text 'Password:' is displayed. The final line shows the prompt 'root@3c3124dec58c:~#' followed by a cursor.

```
student@3c3124dec58c:~$ su - root
Password:
root@3c3124dec58c:~#
```

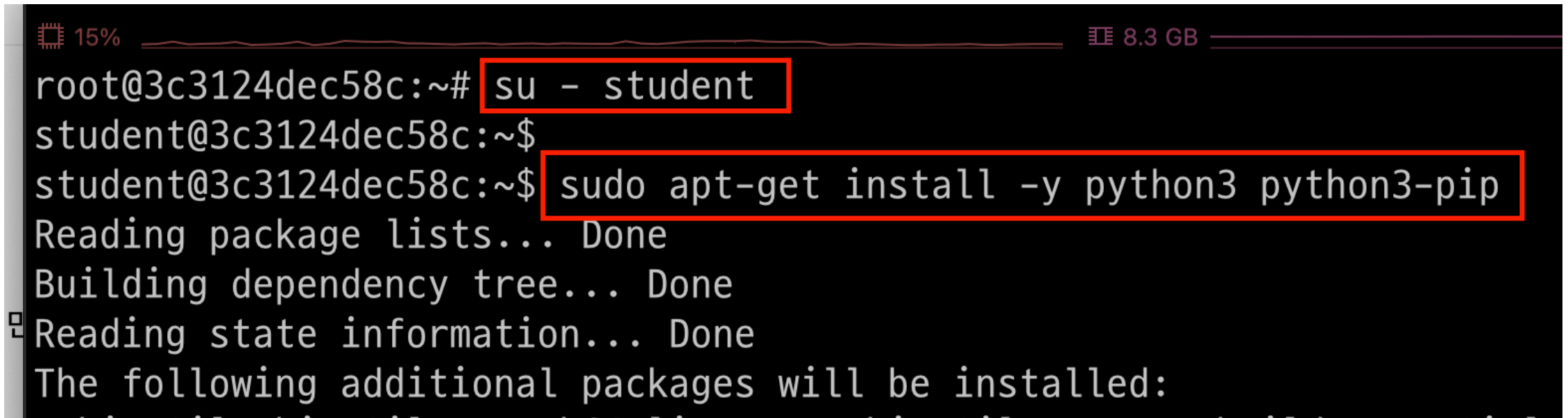
- student 계정에 sudo 권한 부여

```
sudo vim /etc/sudoers  
# 아래내용 추가  
student ALL=(ALL:ALL) ALL
```

```
12% 8.0 GB  
# "sudo scp" or "sudo rsync" should be able to use your SSH agent.  
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"  
  
# Ditto for GPG agent  
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"  
  
# Host alias specification  
  
# User alias specification  
  
# Cmnd alias specification  
  
# User privilege specification  
root    ALL=(ALL:ALL) ALL  
student ALL=(ALL:ALL) ALL  
  
# Members of the admin group may gain root privileges  
%admin  ALL=(ALL) ALL
```

- student 계정으로 sudo 실행

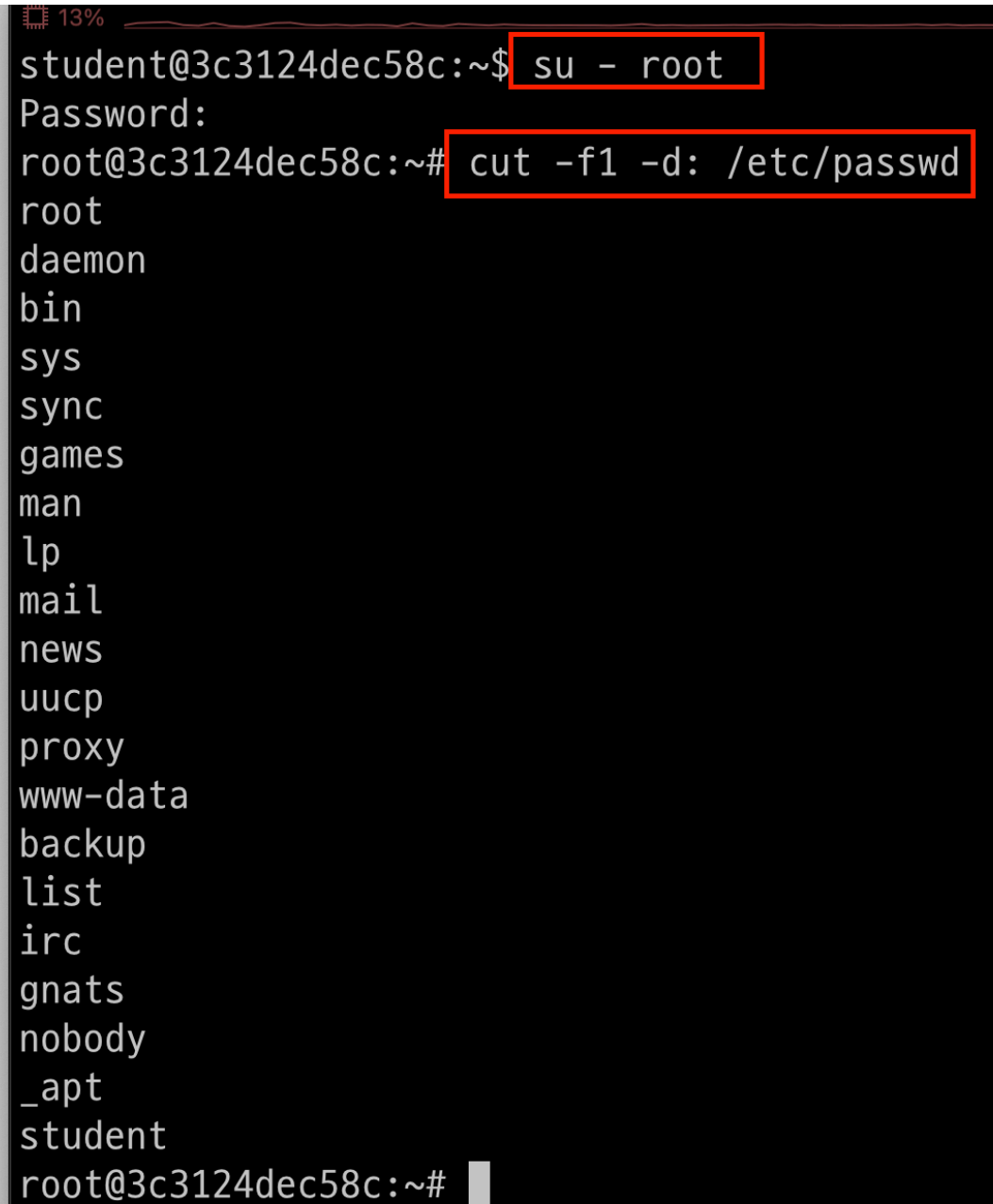
```
su - student # student 계정으로 변경  
sudo apt-get install -y python3 python3-pip
```



```
root@3c3124dec58c:~# su - student  
student@3c3124dec58c:~$  
student@3c3124dec58c:~$ sudo apt-get install -y python3 python3-pip  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:
```

ubuntu 계정 조회

```
su - root  
cut -f1 -d: /etc/passwd
```



A terminal window with a black background and white text. At the top left, there is a battery icon and the text '13%'. The prompt is 'student@3c3124dec58c:~\$'. The user enters 'su - root', which is highlighted with a red box. The prompt changes to 'root@3c3124dec58c:~#'. The user enters 'cut -f1 -d: /etc/passwd', which is also highlighted with a red box. The terminal then displays a list of usernames: root, daemon, bin, sys, sync, games, man, lp, mail, news, uucp, proxy, www-data, backup, list, irc, gnats, nobody, _apt, student. The prompt at the bottom is 'root@3c3124dec58c:~#' followed by a cursor.

```
student@3c3124dec58c:~$ su - root  
Password:  
root@3c3124dec58c:~# cut -f1 -d: /etc/passwd  
root  
daemon  
bin  
sys  
sync  
games  
man  
lp  
mail  
news  
uucp  
proxy  
www-data  
backup  
list  
irc  
gnats  
nobody  
_apt  
student  
root@3c3124dec58c:~#
```

student 계정 삭제

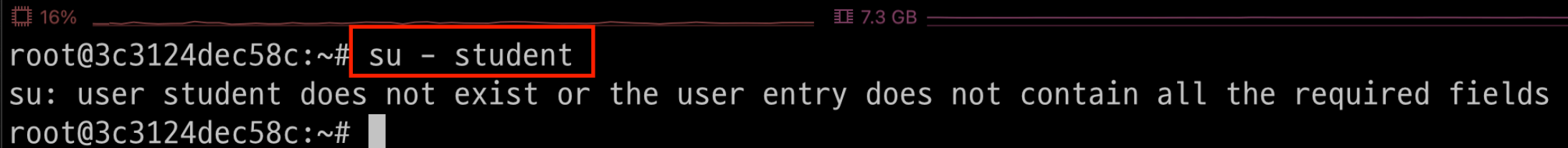
- 명령어: sudo deluser <계정 아이디>

```
sudo deluser student
cut -f1 -d: /etc/passwd # 결과 확인
```

```
12%
root@3c3124dec58c:~# sudo deluser student
Removing user `student' ...
Warning: group `student' has no more members.
Done.
root@3c3124dec58c:~# cut -f1 -d: /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
gnats
nobody
_apt
root@3c3124dec58c:~#
```

- 오류 발생 > student 계정없음

```
su - student
```



A terminal window with a dark background. The top status bar shows '16%' on the left and '7.3 GB' on the right. The prompt is 'root@3c3124dec58c:~#'. The command 'su - student' is entered and highlighted with a red box. The output is 'su: user student does not exist or the user entry does not contain all the required fields'. The prompt returns to 'root@3c3124dec58c:~#' with a cursor.

```
root@3c3124dec58c:~# su - student
su: user student does not exist or the user entry does not contain all the required fields
root@3c3124dec58c:~#
```